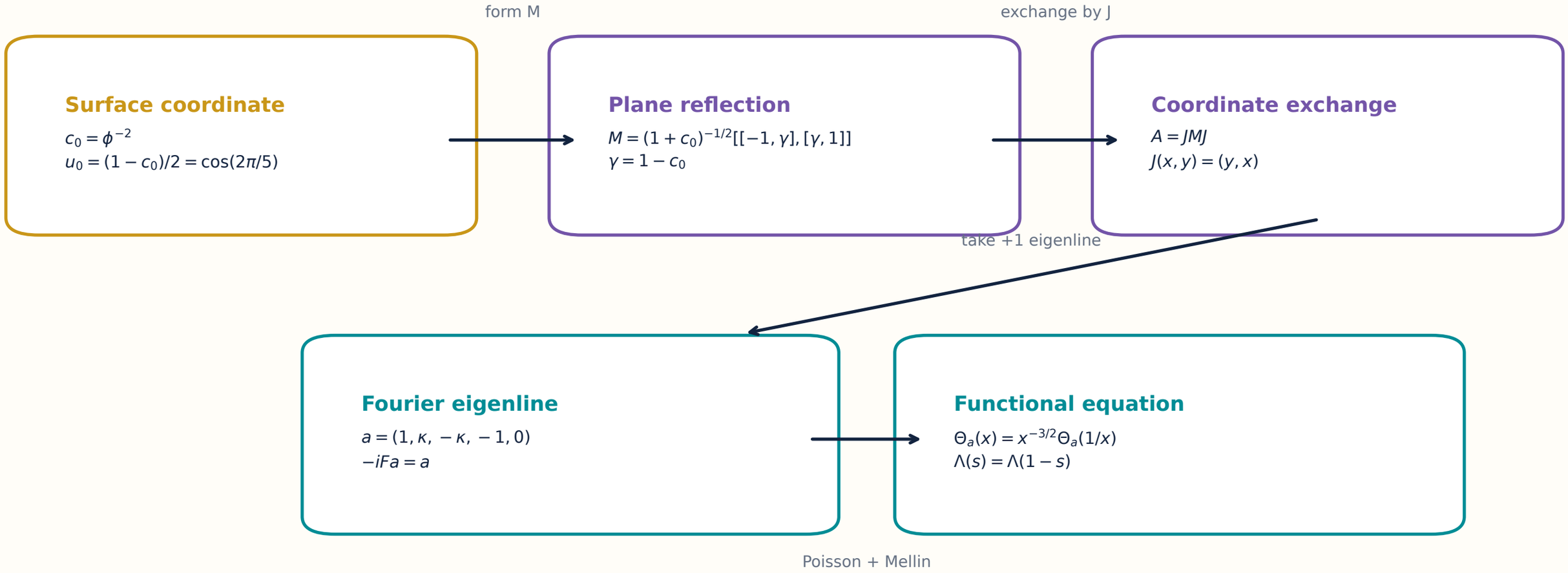

Distinct CM branch: i is fixed by $S : \tau \mapsto -1/\tau$ and selects $\kappa = r(i)$.

$\tau_{163} = (1 + i\sqrt{163})/2$ instead supplies $j(\tau_{163}) = -640320^3$ and a tiny negative nome.